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CH 6: BASIC PROJECT PLANNING TECHNIQUES, PART 2

Prof Corné Schutte

Project Organization and Responsibility



- While creating WBS, the questions “who is needed to do this” and “who will be responsible” are addressed for each package
- Answers result in responsibility assignments for all areas of project
- People responsible and details of that responsibility are documented and communicated in Responsibility Matrix

Responsibility Matrix (RACI)



- For each task, shows who is responsible
- Also called **RACI** Matrix to indicate who should be Responsible, Accountable, Consulted, and Informed
- For each person in matrix, can also show the *kind* of responsibility; e.g.
 - P: primary or lead
 - One, and only one, P per task
 - S: secondary
 - N: notification required
 - A: approval required



P Primary responsibility
S Secondary responsibility
N Must be notified
A Must give approval

[illegible]

Responsibility Matrix - Criteria

- Every task accounted for; nothing falls through cracks
- Each responsibility represents mutual agreement

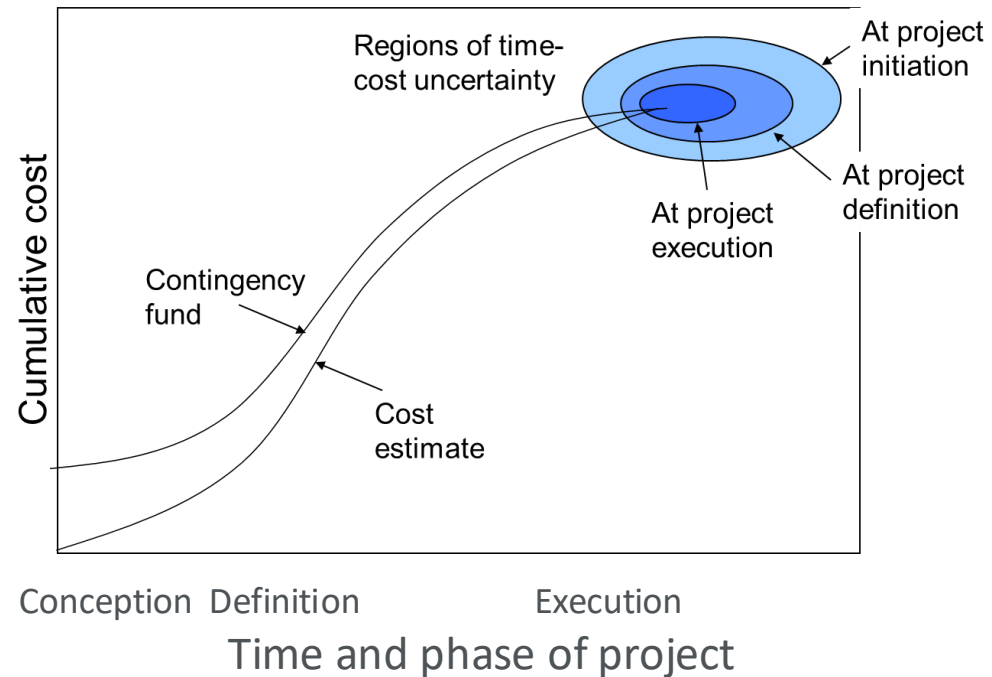


Estimating Time, Costs, and Resources



Estimating in the Project Life Cycle

- Estimates are needed during project conception to approve the project
- They are needed during project definition to create the project schedule and budget
- Early estimates are sometimes uncertain or inaccurate due to lack of information
- Later estimates are more accurate.
- Uncertainty of estimates is covered by “contingencies” or “buffers”



Estimating Process



Estimate versus Target or Goal

- **Estimate:** a *realistic assessment* based upon known facts about the work, required resources, constraints, and the environment, derived from estimating methods
- **Target or goal:** a desired outcome, commitment, or promise.
- Don't confuse estimates with goals. The estimating process is directed at producing good estimates, not restating targets or goals.

Estimating Process



Accuracy versus Precision

- **Accuracy:** the closeness of the estimated value to the actual value
- **Precision:** the number of decimal places in the estimate.
- Accuracy of estimates is more important than precision

Estimating Process



Classification of Work and Costs

- **Off-the Shelf (OTS):** Adaptation, repetition, or reuse of something that already exists.
- **Developmental:** New, innovation, or non-existing. Start from scratch.

Estimating is much more difficult for developmental than OTS.

- **Recurring:** happen repeatedly during project.
- **Non-recurring:** happen once.

Estimating Process

Estimating Methods

1. Expert judgement
2. Analogy + compensation for differences
3. Parametric: Formula or Cost Function, e.g.,

Cost, engine A = (Cost, engine B)

$$\left(\frac{\text{Thrust, engine A}}{\text{Thrust, engine B}} \right)^{0.7}$$

Cost, cabling = 150 (total area + 10%)
+ 300 (number of rooms)
+ 125 (number of floors)



Time estimate, hours = 3 + 0.25(number units installed)

Estimating Methods (cont'd)

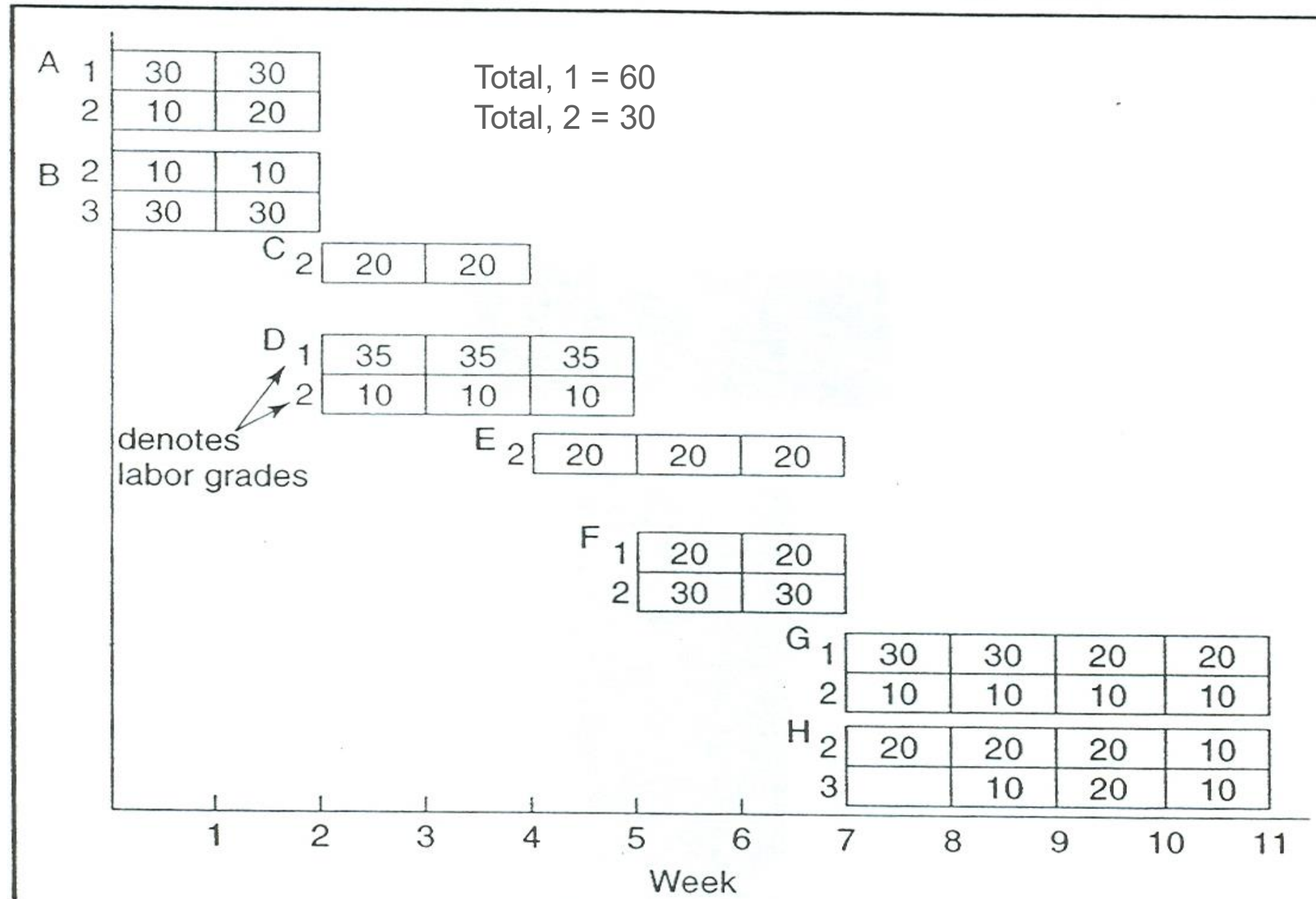


4. Cost engineering

Detailed cost breakdown of labor, materials, etc. at the work package or task level.

Example below

Schedule showing hours allocated to work packages by labor grade.



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Labor Hours and Nonlabor Costs.

WORK PACKAGE	HOURS BY LABOR GRADE			NONLABOR COSTS			
	1	2	3	MATERIAL	EQUIPMENT	SUBCONTRACTS	OTHER
A	60	30		\$ 500			
B		20	60		\$1,000		
C		40			500		\$ 500
D	105	30			500		
E		60				\$4,500	
F	40	60		8,000	1,000	5,000	500
G	100	40		1,500			500
H		70	40		1,000		1,500
Total	305	350	100	\$10,000	\$4,000	\$9,500	\$3,000

Labor Grade **Hours x Labor Rate x (100% + Overhead Rate) = Labor Cost**

1	305 (\$20)(100% + 90%) = \$11,590
2	350 (\$25)(100% + 100%) = \$17,500
3	100 (\$40)(100% + 120%) = <u>\$8,800</u>
	<u>\$37,890</u>

Preliminary Estimate = Labor Cost + Nonlabor Costs

$$\begin{aligned}
 &= \$37,890 + \$26,500 \\
 &= \$64,390
 \end{aligned}$$

Total = \$26,500

Final Estimate = Preliminary + G/A Rate

$$\begin{aligned}
 &= \$64,390 (100\% + 10\%) \\
 &= \$70,829
 \end{aligned}$$

Budget = Final Estimate + Contingency (Reserve)

$$\begin{aligned}
 &= \$70,829 + \$7,100 \\
 &= \$77,929
 \end{aligned}$$



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Estimating Methods (cont'd)



5. Delphi

- Application of expert judgement that aggregates judgements of several people.
- Utilizes iterative series of questionnaires to “home in” on most widely-shared opinion of estimates

Estimating Process

- Any of these methods can be used in any area of project
- Parametric and cost engineering methods are the best when the data exists to use them

Estimating Process



Contingency Fund

aka, Contingency Reserve or Management Reserve

- Amount added to the budget or schedule to account for uncertainty in estimates.
- Can be added to budget/schedule for individual activities or entire project.
- For schedules, called “Schedule Reserve” or “Time Buffer”
- As project progresses and uncertainty declines, the fund or reserve is reduced.

Estimating Process

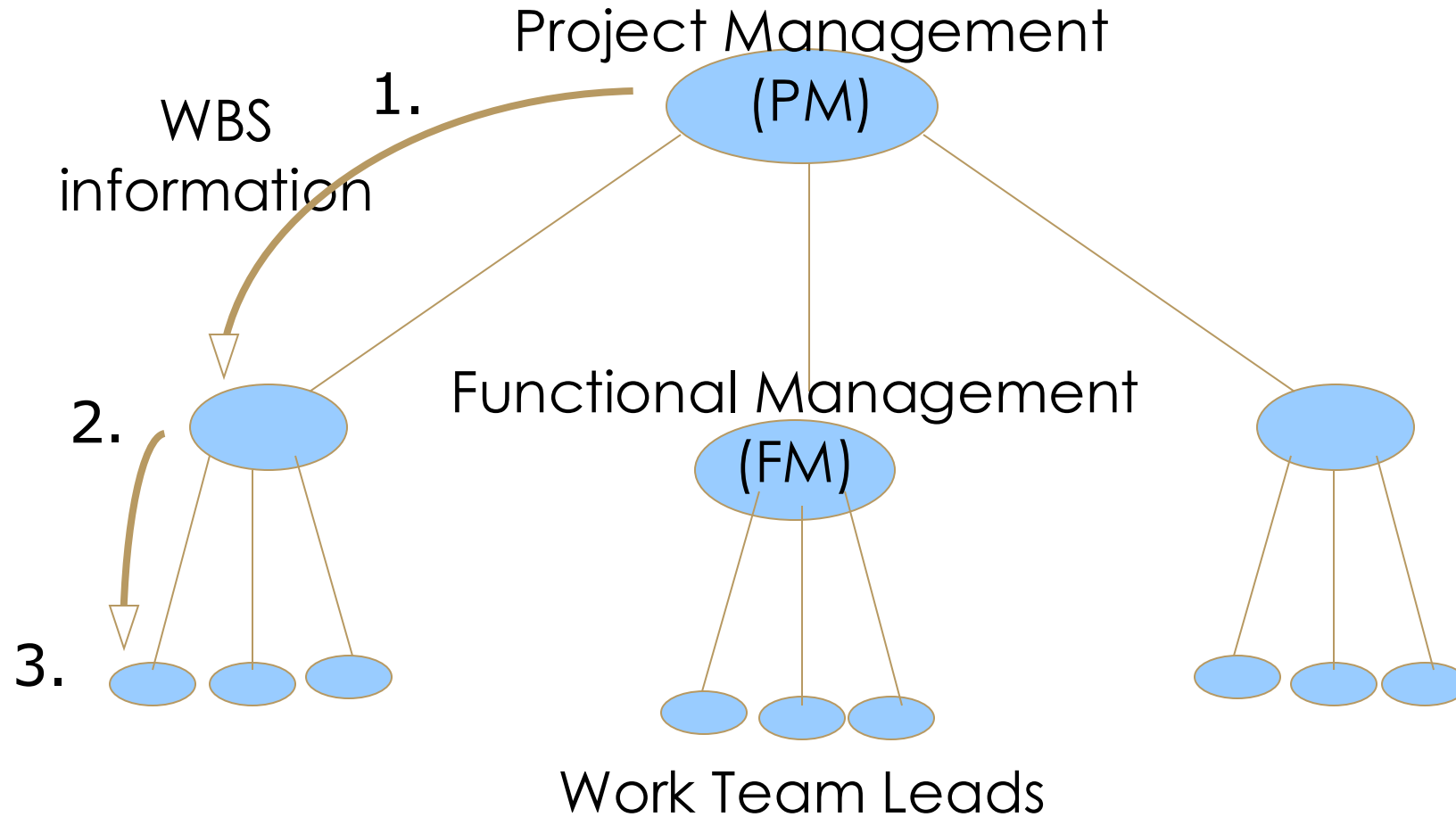


- Rule of Thumb:

The smaller the work packages or portion of the end-item estimated, the better the estimate

Estimating Process

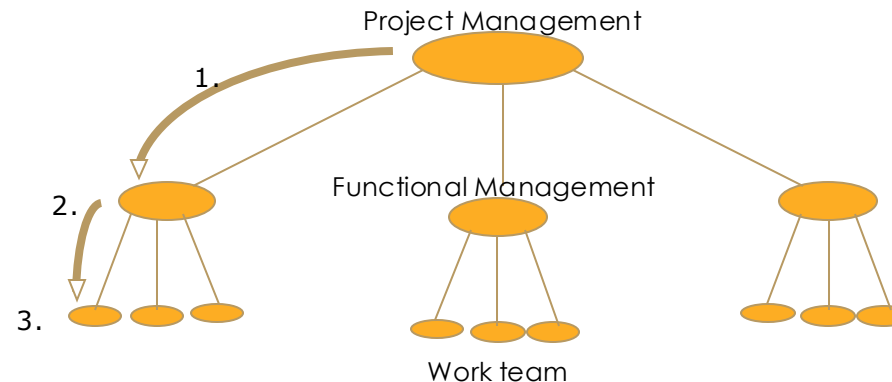
- For small, simple projects, the project manager alone might do all the estimating.
- For larger projects, use “bottom-up” procedure, described below



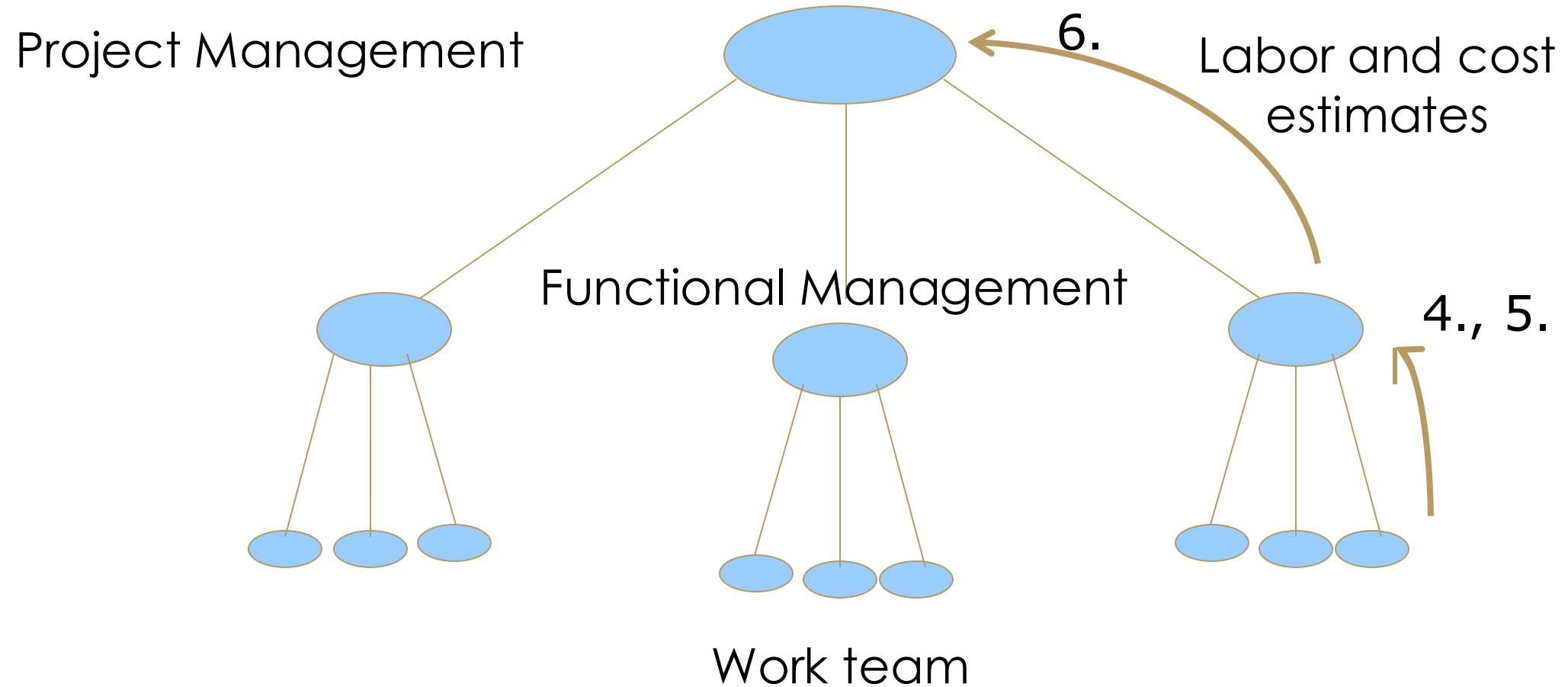
Estimating Process

Procedure for larger projects, steps 1-3

1. PM: Uses WBS to identify work packages
2. FM: Subdivide work packages into identifiable tasks; determine labor, material, facilities, and resources requirements for each
3. Supervisors/team leads: Estimate number of labor hours and quantities of materials needed



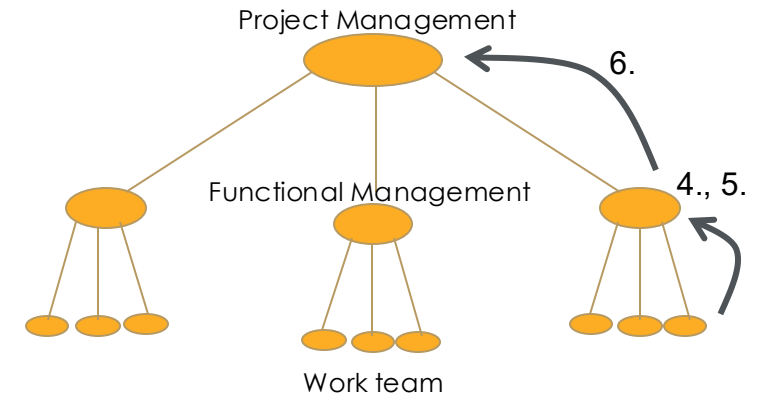
Estimating Process



Estimating Process

Procedure for larger projects, steps 4-6

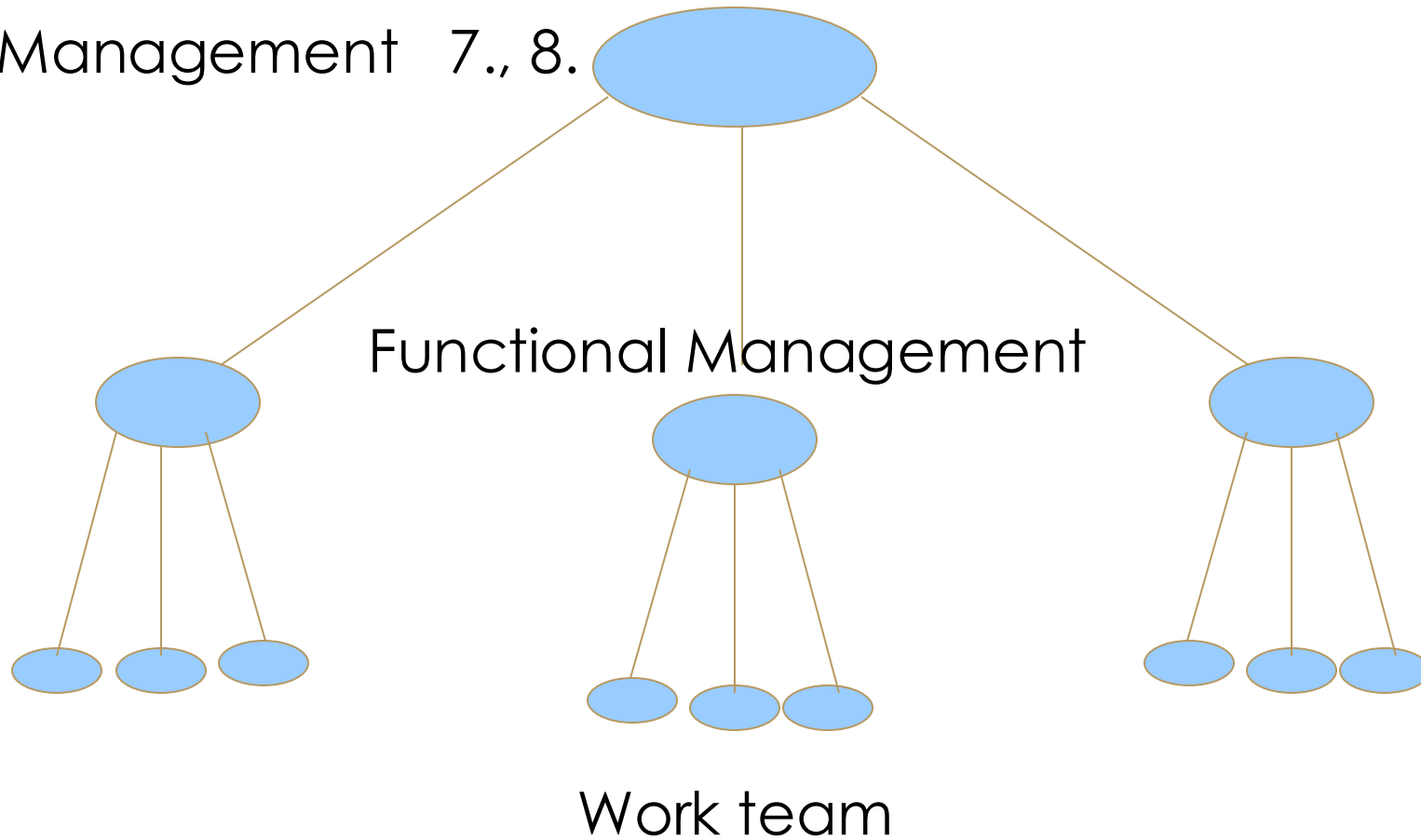
4. FM: check and aggregate time and material estimates
5. FM: convert time estimates into costs
6. PM: checks over and approves all estimates aggregates costs; added in overhead costs:



$$\text{Project cost} = \sum \text{direct costs} + \sum \text{overhead costs}$$

Estimating Process

Project Management 7., 8.



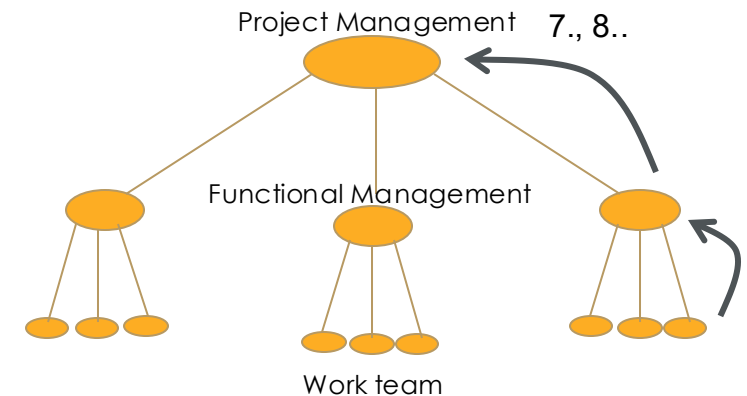
Estimating Process

Procedure for larger projects, steps 7-8

7. PM: Adds in contingency amounts.

Two possible contingencies

1. Base estimate = Σ (WP estimates + WP contingency)
(to handle “known-unknowns”)
 2. Final estimate = Base estimate + overheads + project contingency
(to handle “unknown unknowns”; PM controls this)
8. PM: Compares bottom-up estimates to top-down targets or goals. Attempt to reconcile differences.



Estimating Process



Estimates can be made at any level

- Project
- Sub-project
- Work package
- Task / Activity

Project Scheduling, the Basics

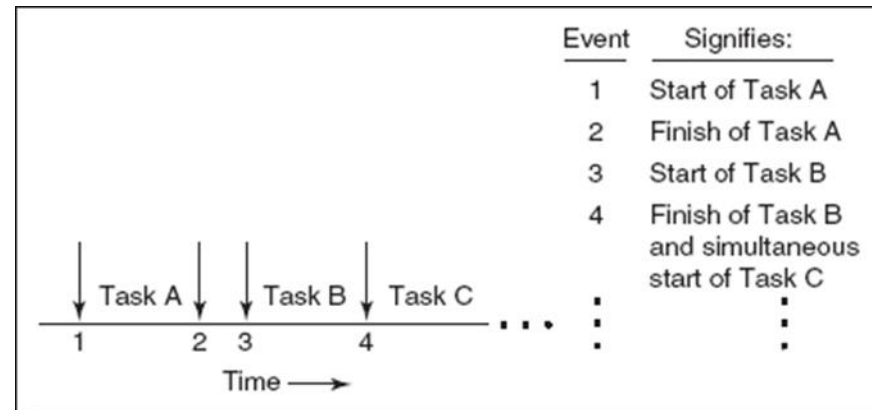


- WBS provides information necessary to create a schedule
- Includes
 - List of tasks (work packages)
 - For each task
 - Duration (or target completion date)
 - Resources required
 - Inputs, preconditions, prior completed tasks (Logical sequencing)
 - Events and milestones measure progress

Tasks and Events

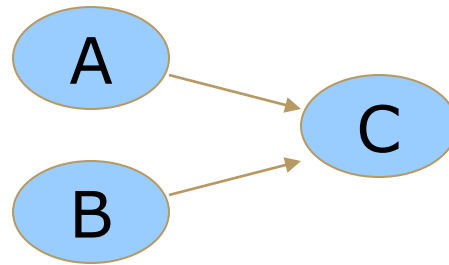
Task represents work effort and consumes resources and time

Event signifies a moment in time, such as the start or finish of a task. A **Milestone** is an event of some significance.



Logical Sequencing of Tasks

- Predecessor: A task that must be completed before another can be started
- If task C depends upon Tasks A and B, then Tasks A and B are “predecessors” for Task C.



- Task C is the “successor” of Tasks A and B

Gantt Chart

- Simple, common scheduling tool
- Easy to create and understand
- Developed by Henry Gantt, a consultant of Frederick Taylor.



H. Gantt

<http://www.ganttchart.com/histor50.png>

- From Wikipedia

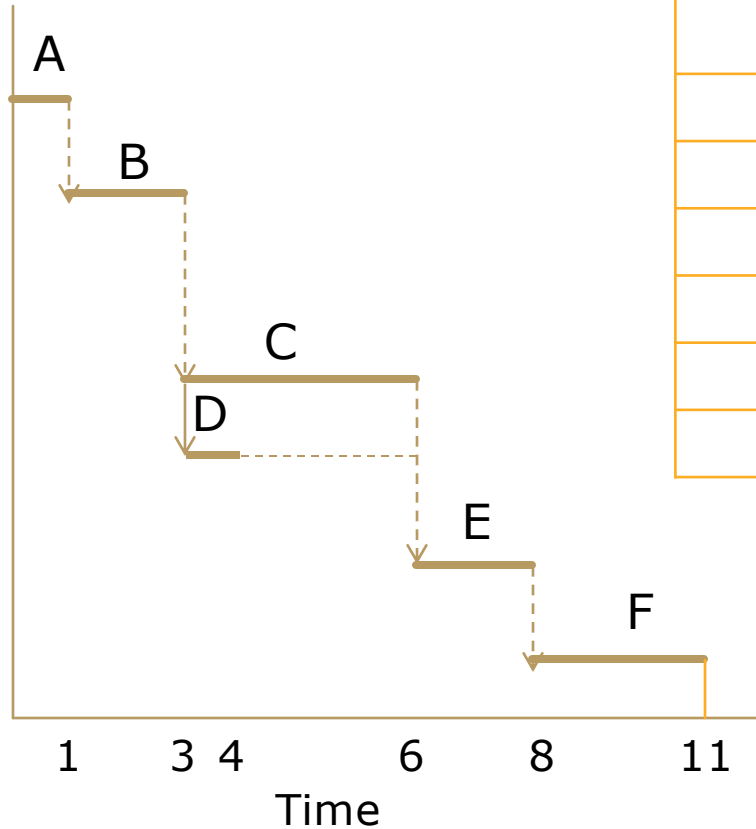
Henry Laurence Gantt, A.B., M.E. (1861-1919) was a mechanical engineer and management consultant who is most famous for developing the Gantt chart in the 1910s. These Gantt charts were employed on major infrastructure projects including the Hoover Dam and Interstate highway system and still are an important tool in project management.

Gantt Chart

- Example

Activity	Immediate Predecessors	Time
A	--	1
B	A	2
C	B	3
D	B	1
E	C, D	2
F	E	3

Gantt Chart



Activity	Immediate Predecessors	Time
A	--	1
B	A	2
C	B	3
D	B	1
E	C, D	2
F	E	3

- Note:

E starts at time 6 because then both C and D are completed. D is completed at time 4, but C is not until time 6. Hence, C must wait until time 6.

Gantt Chart



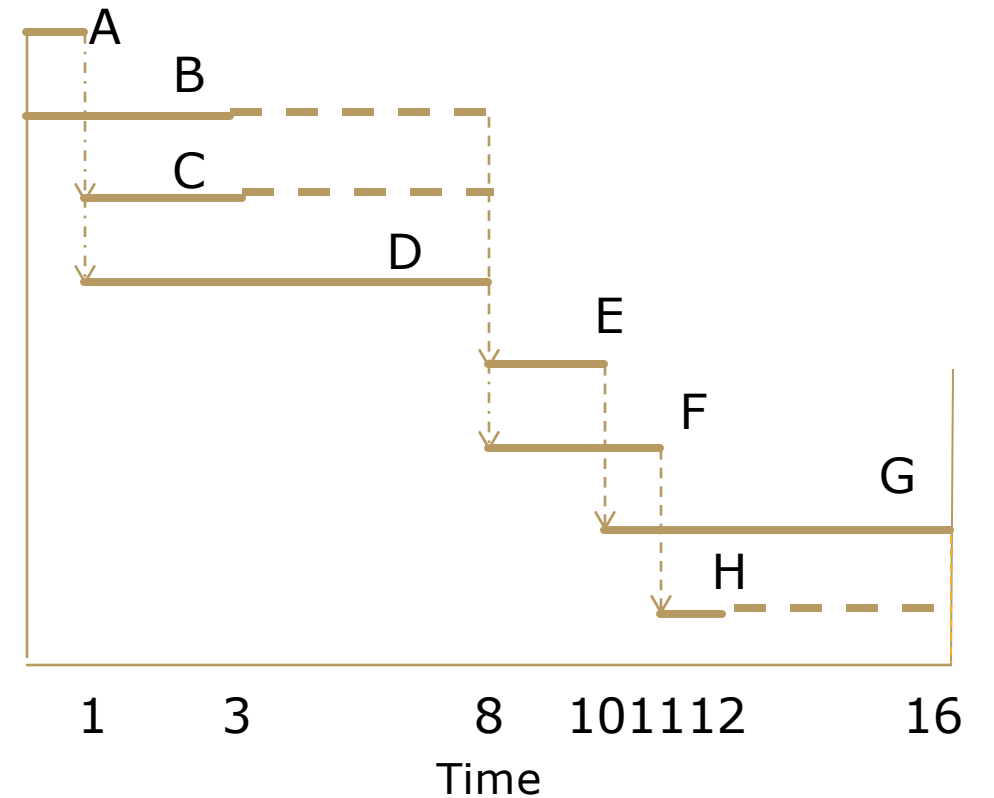
- Another Example

Activity	Immediate Predecessors	Time
A	--	1
B	--	3
C	A	2
D	A	7
E	B, C, D	2
F	B, C, D	3
G	E	6
H	F	1

Gantt Chart

Activity	Immediate Predecessors	Time
A	--	1
B	--	3
C	A	2
D	A	7
E	B, C, D	2
F	B, C, D	3
G	E	6
H	F	1

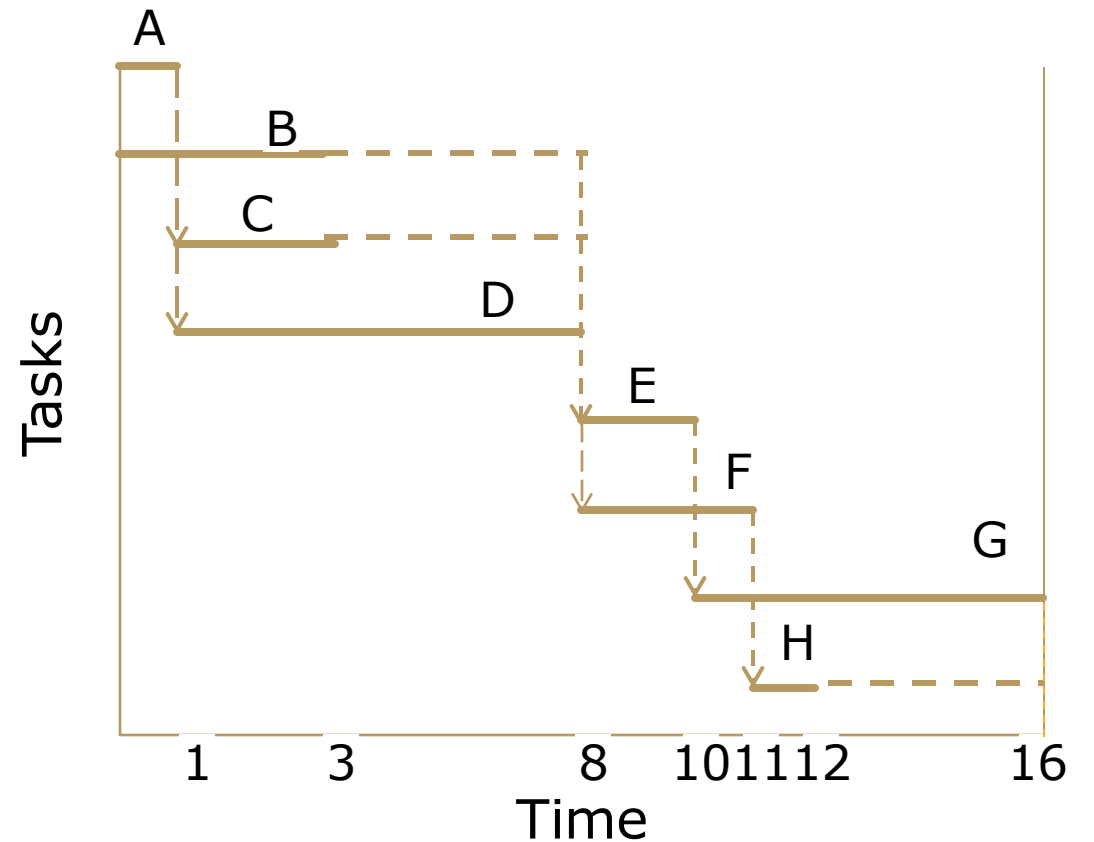
Tasks



Gantt Chart

Notes:

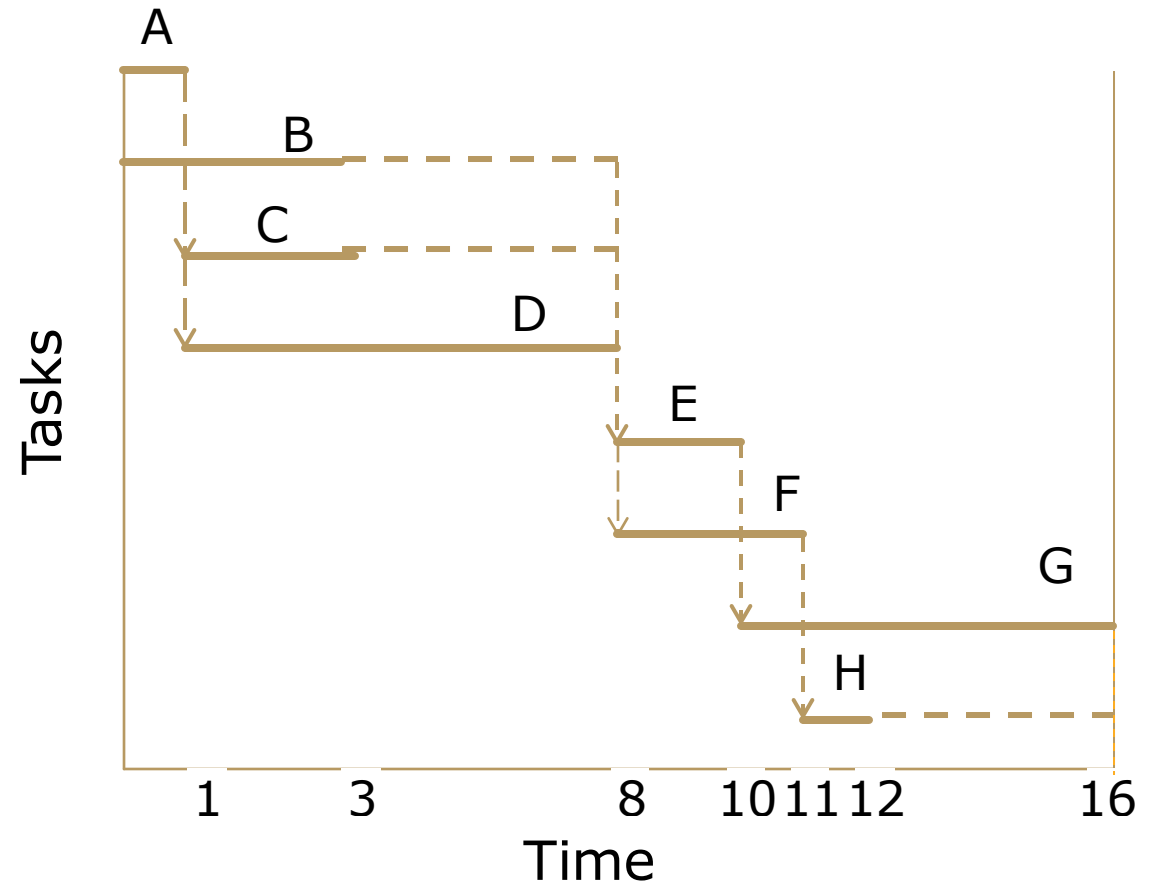
- Dotted lines show predecessor relationships
- Typically, x-axis of Gantt chart shows calendar dates and includes time off for weekends and holidays. Example shows only elapsed times.



Gantt Chart

Notes (cont'd):

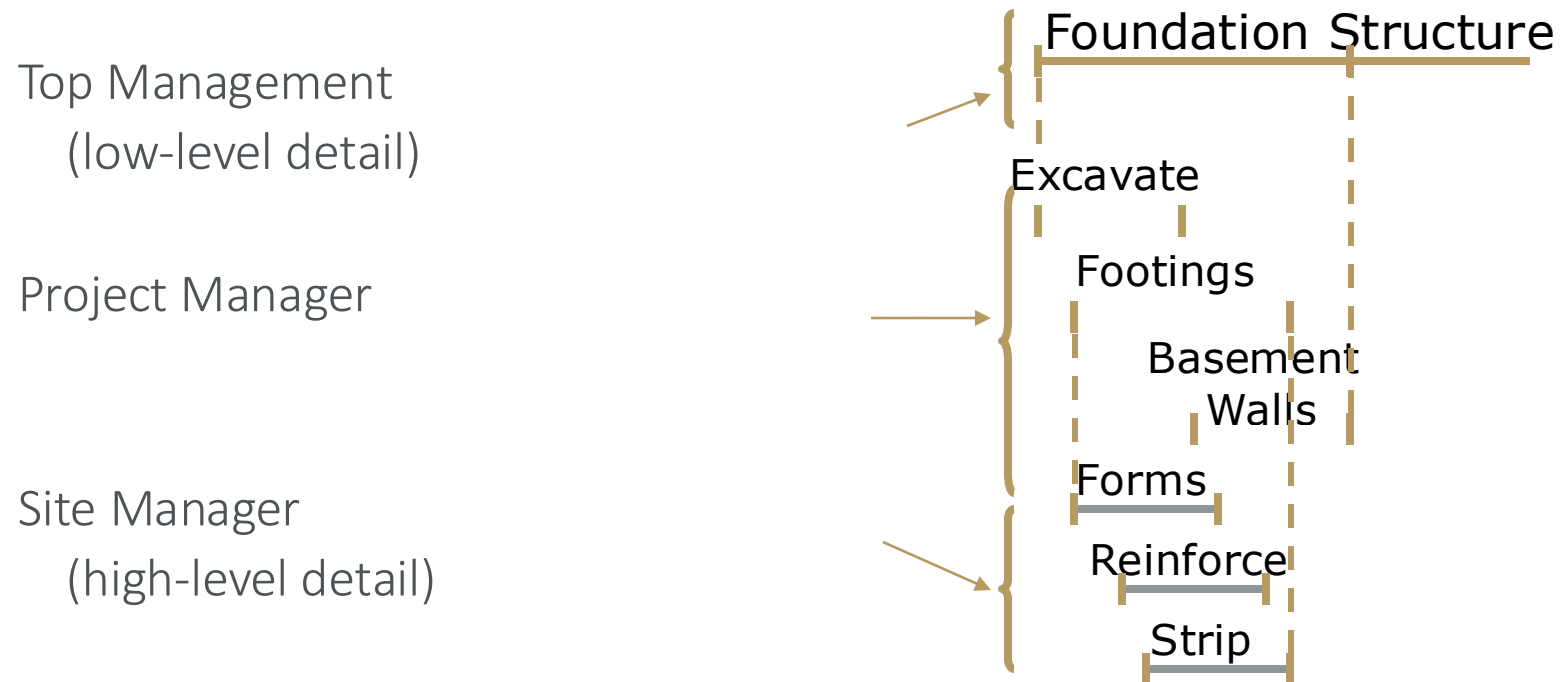
- This project is completed after 16 weeks
- This example shows only “events” or points in time. No allowance made for weekends or other time off



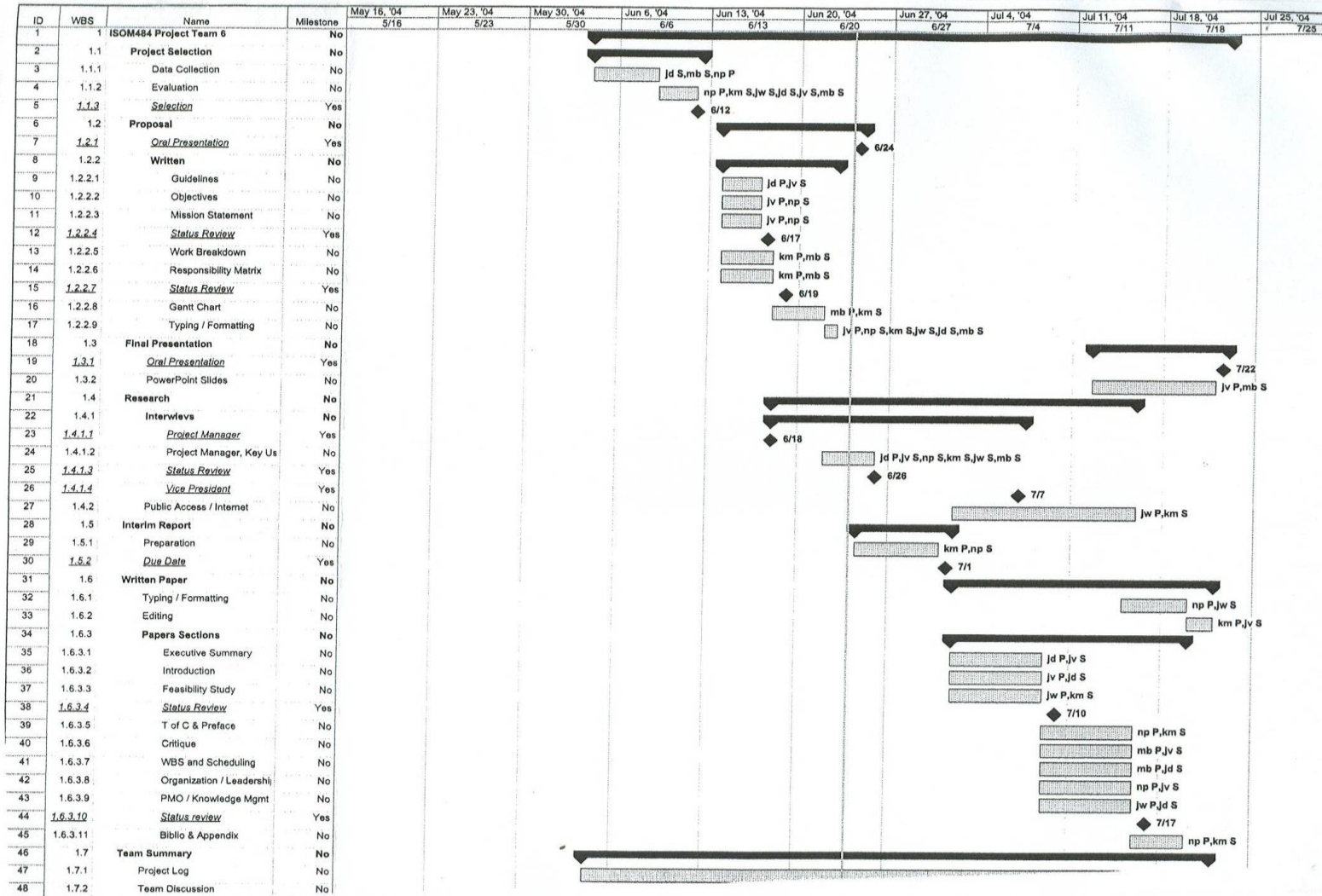
Hierarchy of Schedules

Scope and Level of detail in the schedule should reflect the audience

- Example

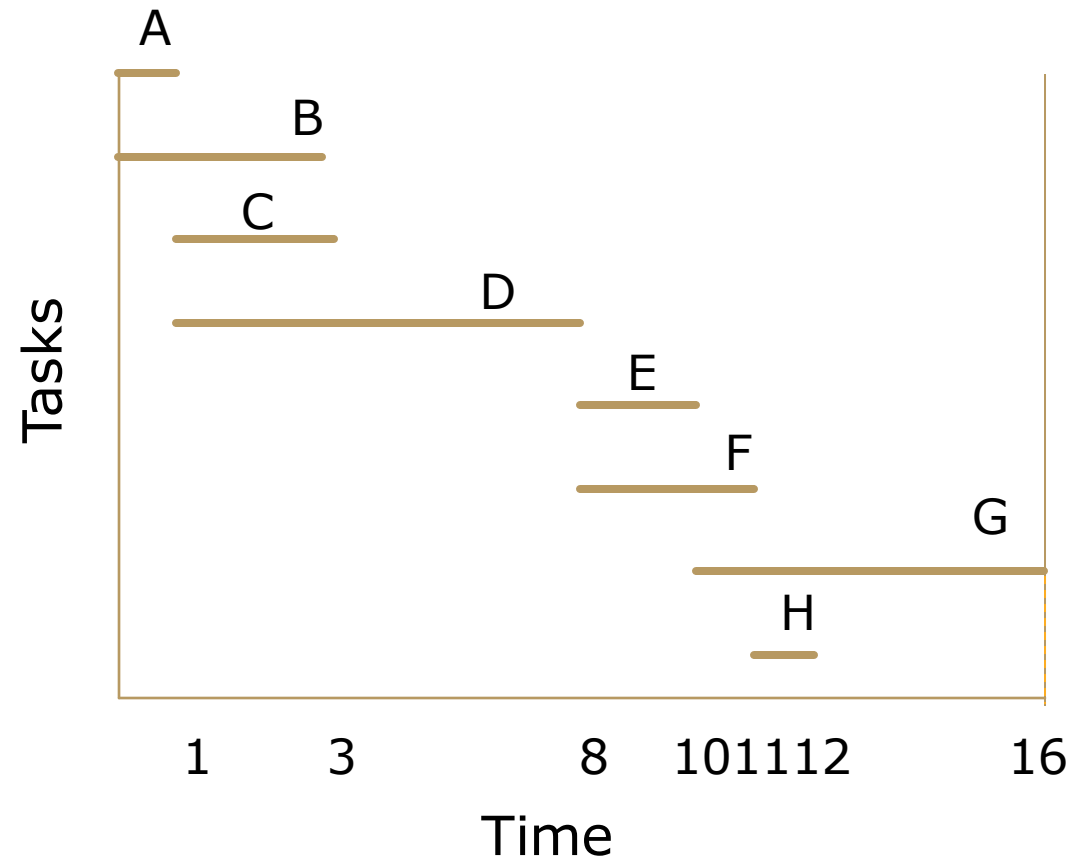


Level of Detail



Gantt Chart: Pro and Con

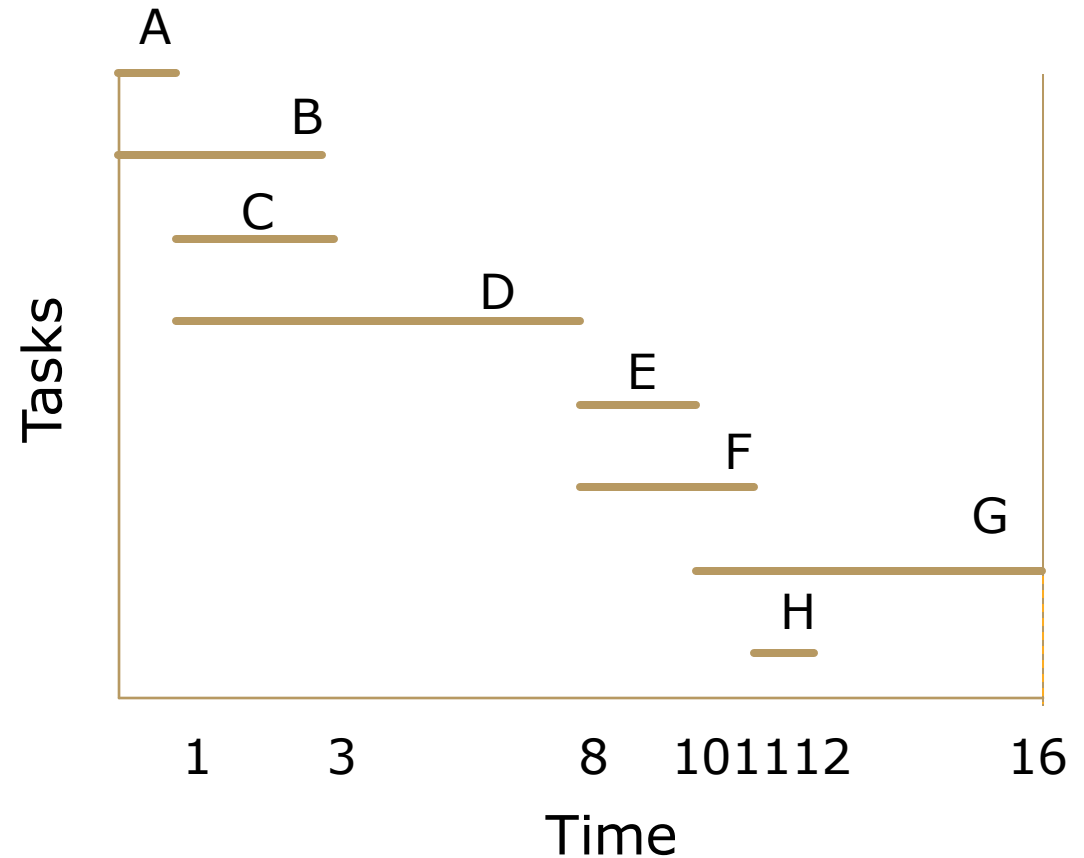
- Pro: Easy to construct and understand
- Con: does not necessarily indicate relationships among tasks, so is limited as tool for planning and control.
- Example:



Gantt Chart: Pro and Con

What is effect of

- C starting 2 weeks late?
- C starting 6 weeks late?
- E taking 3 weeks instead of 2 weeks?
- E finishing a week early?



Gantt Chart: Pros and Cons

- With simple Gantt charts, such questions are not always easy to answer.
- Yet you need the answers to plan the project and create realistic schedules
- Gantt charts are good for displaying schedules,
 - Networks, described in the next chapter, are better for creating them

Thank you
Enkosi
Dankie

